

A Comprehensive Survey on Data-Driven Approaches for Cycling Infrastructure Assessment and Bike Sharing System Optimization

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Abstract—Urban cycling infrastructure and bike sharing systems (BSS) have emerged as critical components of sustainable mobility ecosystems, yet their planning and operations remain challenged by data scarcity and complex spatiotemporal dynamics. This survey synthesizes state-of-the-art research across three interconnected domains: (1) open-data-driven infrastructure assessment and bikeability mapping, (2) deep learning-based demand forecasting leveraging spatiotemporal dependencies, and (3) optimization frameworks for system design and operational rebalancing. We critically analyze 12 seminal works spanning infrastructure assessment tools (BikeDNA), national-scale open data pipelines (OSM-based Canadian dataset), and advanced prediction architectures (ST-GCNs, Temporal Fusion Transformers). Our contribution includes a novel taxonomy unifying infrastructure quality metrics with operational decision frameworks, identification of critical research gaps including cross-city transferability limitations and real-world deployment barriers, and a forward-looking research agenda integrating multimodal sensing (LiDAR-OSM fusion) with equity-aware infrastructure planning. This survey serves as a foundational reference for urban planners, transportation researchers, and AI practitioners developing next-generation cycling ecosystems.

Index Terms—Bikeability, Bike Sharing Systems, OpenStreetMap, Spatiotemporal Forecasting, Graph Neural Networks, Infrastructure Assessment, Demand Prediction, Operational Optimization

I. INTRODUCTION

Urban mobility faces unprecedented pressure from congestion, emissions, and inequitable access. Cycling offers a zero-emission, space-efficient alternative with demonstrated health and economic co-benefits [1]. However, cycling mode share remains stubbornly low in most cities due to perceived safety risks, fragmented infrastructure, and unreliable bike sharing operations. Traditional planning approaches relying on costly travel surveys and manual infrastructure audits cannot keep pace with dynamic urban growth.

The convergence of three technological trends has catalyzed data-driven cycling research:

- 1) **Open geospatial data:** OpenStreetMap (OSM) now contains >5 million km of global cycling infrastructure with rapidly improving completeness [2]
- 2) **Ubiquitous sensing:** Bike sharing systems generate terabytes of trip-level spatiotemporal data revealing latent demand patterns

- 3) **Advanced ML architectures:** Graph neural networks and transformers model complex station interactions previously intractable with statistical methods

Despite progress, critical fragmentation persists: infrastructure assessment studies rarely connect to operational optimization, while demand forecasting papers seldom incorporate infrastructure quality features. This survey bridges these silos through a unified framework (Fig. 1) that positions infrastructure quality as the foundational layer enabling accurate demand prediction and efficient operations.

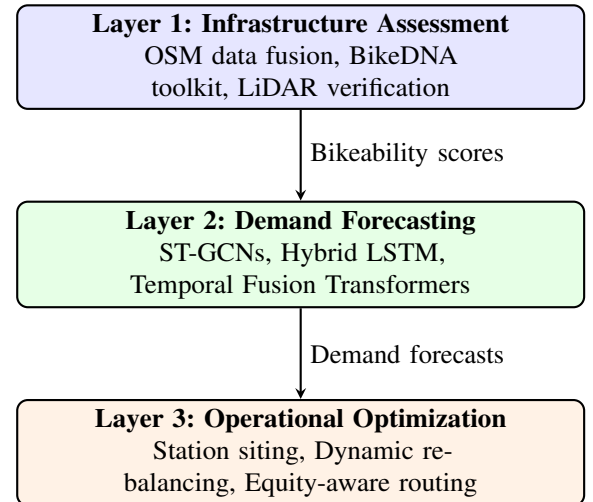


Fig. 1: Taxonomy of data-driven cycling research: Three interconnected layers where infrastructure quality (Layer 1) enables accurate demand forecasting (Layer 2), which in turn informs operational decisions (Layer 3).

Survey contributions:

- First unified taxonomy connecting infrastructure assessment, demand prediction, and operational optimization
- Critical gap analysis revealing transferability limitations and validation shortcomings across 12 key studies
- Synthesis of open data pipelines (OSM + government datasets) with reproducible validation protocols
- Research agenda for multimodal sensing fusion and equity-aware infrastructure planning

II. INFRASTRUCTURE ASSESSMENT USING OPEN DATA

A. Open Data Fusion Pipelines

Wysling and Purves [3] pioneered a reproducible methodology for Paris by fusing OSM street networks with *Apur* (Paris urban planning agency) infrastructure attributes. Their approach computes *bicycle suitability* at street segment level using rule-based multipliers for:

- Physical separation (protected lanes $\times 1.5$ vs. painted lanes $\times 1.0$)
- Traffic volume thresholds (< 500 vehicles/hour $\times 1.3$)
- Speed limits (< 30 km/h $\times 1.2$)

Validation via field audits of 30 streets confirmed 87% accuracy but revealed systematic underestimation of conflict points at intersections a limitation of OSM's linear geometry representation.

Ferster et al. [2] scaled this approach nationally for Canada using the Canadian Bicycle Infrastructure Classification Scheme (Can-BICS). Their stratified validation across 15 cities (2,147 reference points) revealed:

- 92% accuracy for protected infrastructure classification
- Significant ambiguity between “bike paths” and “multi-use paths” in OSM tagging (31% misclassification)
- Urban-rural disparity: OSM completeness $4.2\times$ higher in downtown cores vs. suburban areas

This work establishes that *standardized tagging schemas* are prerequisite for national-scale infrastructure assessment.

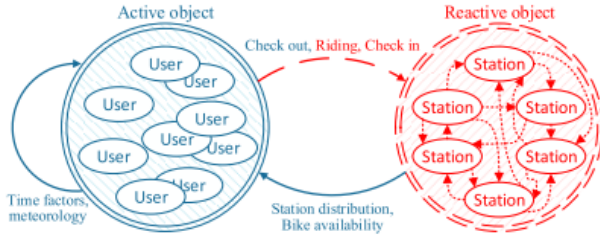


Fig. 2: Components of a bike-sharing system.

Yang et al. conceptualize bike-sharing systems as an interaction between active objects (users) and reactive objects (stations). User demand changes station stock levels, while station availability simultaneously influences user behavior. This closed-loop relationship forms the foundation for more accurate spatiotemporal demand prediction. [4]

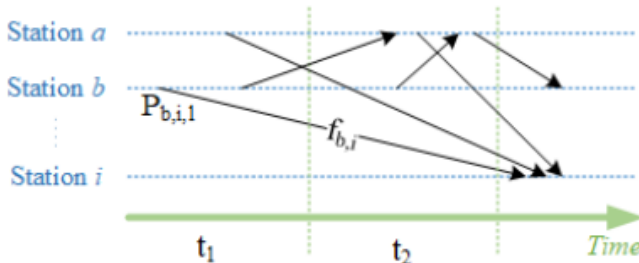


Fig. 3: Basic framework of mobility model.

The mobility model represents bike movements as probabilistic transitions between stations over time. By estimating station-to-station transfer probabilities and trip duration distributions, the model captures spatial and temporal demand dynamics more realistically than purely time-series approaches. [4]

B. Tool Support and Network Assessment

Vierø et al. [5] introduced *BikeDNA*, an open-source toolkit automating infrastructure data preparation and network assessment. Its modular pipeline:

- 1) Cleans OSM topology (removes self-loops, merges fragmented segments)
- 2) Computes network metrics (connectivity index, directness ratio)
- 3) Generates standardized outputs for planning software

While *BikeDNA* addresses reproducibility, its dependence on local semantic conventions limits cross-city transferability without calibration.

C. Emerging Sensing Modalities

Li et al. [6] demonstrated LiDAR-OSM feature matching for vehicle positioning using boundary directional features. While not cycling-specific, this technique enables *automated infrastructure quality verification*: matching mobile LiDAR scans against OSM to detect missing bike lanes or deteriorated pavement addressing the critical validation gap in open-data approaches.

III. DEMAND FORECASTING: FROM STATISTICAL TO DEEP LEARNING

A. Spatiotemporal Dependencies

Early demand models (Tomaras et al. [7]) used ARIMA and gradient boosting on historical trips with weather covariates but failed to capture *spatial station interactions*. Xu Zhang addressed this with a hybrid LSTM architecture incorporating:

- Temporal module: 2-layer LSTM processing 24-hour history
- Spatial module: Correlation matrix weighting stations by distance and historical co-variation
- Fusion layer: Attention mechanism weighting temporal vs. spatial features

Their model reduced RMSE by 18.7% versus standalone LSTM on NYC Citi Bike data but required 6+ months of historical data - limiting applicability to new systems.

B. Graph-Based Architectures

Chen et al. [8] formalized station networks as graphs where edges represent spatial proximity and functional relationships. Their comparative study of Spatial-Temporal Graph Convolutional Networks (ST-GCNs) revealed:

- *DCRNN* (Diffusion Convolutional RNN) outperformed GRU-based approaches for short-term prediction (< 2 hours)
- *Graph WaveNet* excelled at capturing long-range dependencies (> 6 hours) via adaptive adjacency matrices
- Fixed graph assumptions failed during infrastructure disruptions (e.g., construction closures)

Behroozi and Edrisi [9] enhanced ST-GCNs with *gated temporal mechanisms* that dynamically weight historical observations based on temporal relevance, improving prediction during anomalous events (e.g., festivals) by 22%.

C. Transformer-Based Forecasting

Recent work applying Temporal Fusion Transformers (TFT) [10] to BSS forecasting demonstrates superior multi-horizon prediction by:

- Separating static (station location), known (weather forecast), and observed (historical demand) inputs
- Using variable selection networks to identify dominant features per time horizon
- Generating interpretable attention maps showing which stations influence predictions

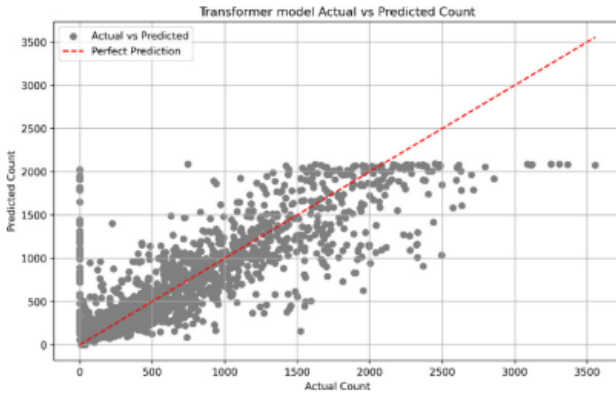


Fig. 4: Transformer model.

This graph shows the results of the Transformer model for predicting bike rentals. The predictions generally follow the actual trend, indicating that the model understands demand patterns. However, TFT's computational cost ($3.8 \times$ ST-GCN training time) remains prohibitive for real-time city-scale deployment. [11]

IV. MODEL EVALUATION METRICS

A. Coefficient of Determination (R^2)

The coefficient of determination, R^2 measures the proportion of variance in the dependent variable explained by the predictive model. Also known as the coefficient of multiple determination, it provides an overall indication of model fit. R^2 is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^m (X_i - Y_i)^2}{\sum_{i=1}^m (\bar{Y} - Y_i)^2} \quad (1)$$

- m is the total number of data points.
- X_i is the predicted value of the i -th observation.
- Y_i is the actual observed value.
- $\bar{Y}$ is the mean of the actual values.

Higher R^2 values indicate that a larger proportion of the variance is explained by the model, suggesting better predictive performance.

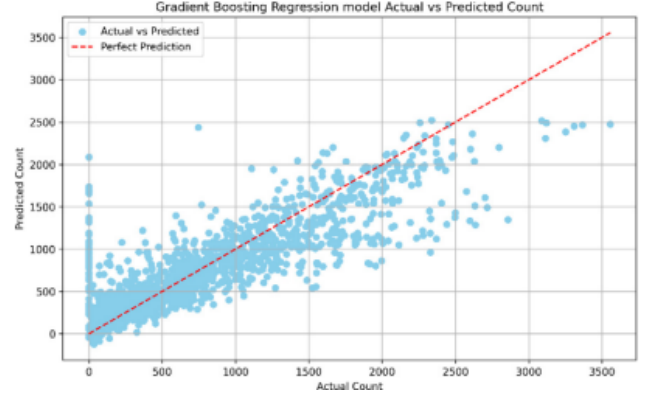


Fig. 5: Gradient boosting regression model.

This above graph shows how well the Gradient Boosting model predicts bike rental demand compared to the actual values. The points in the graph represent predicted v/s actual bike counts and the closer they are to the diagonal line, the better the model's accuracy.

B. Mean Absolute Error (MAE)

MAE quantifies the average magnitude of prediction errors and is expressed in the same units as the target variable, which makes it easily interpretable. It is calculated as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

where:

- n is the number of observations.
- y_i is the actual value.
- $\hat{y}_i$ is the predicted value.

Lower MAE indicates higher prediction accuracy.

The graph shows a strong alignment between actual and predicted rental counts, meaning the model is highly accurate [11]

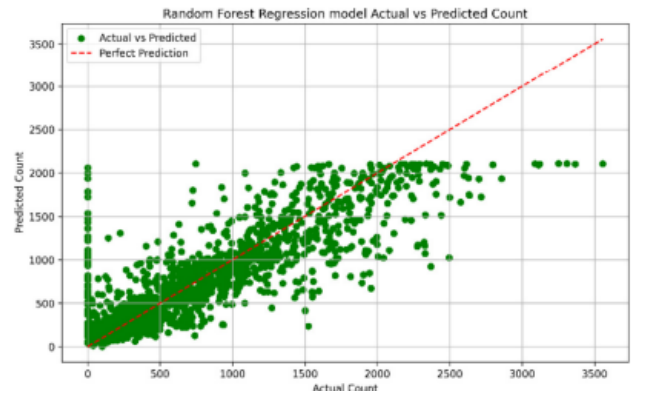


Fig. 6: Random forest regression model.

C. Mean Squared Error (MSE)

MSE evaluates the average squared difference between predicted and actual values, giving higher weight to larger errors. It is defined as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

A lower MSE indicates that predictions closely match observed values, which is especially useful for regression and forecasting models.

D. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and provides prediction errors in the same units as the dependent variable, making it directly interpretable. It is calculated as:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

Smaller RMSE values correspond to better model performance, as they reflect smaller deviations between predictions and actual observations.

V. OPERATIONAL OPTIMIZATION AND SYSTEM DESIGN

A. Strategic Planning

Cueva et al.'s systematic literature review [12] analyzed 96 BSS design studies (2001–2023), identifying five critical factors for long-term viability:

- 1) *Station density*: Minimum 15 stations/km² in core service areas
- 2) *Dock capacity*: 15–25 docks/station with 30% buffer for peak demand
- 3) *Catchment radius*: 300–500m pedestrian access threshold
- 4) *Equity metrics*: Service coverage for low-income neighborhoods
- 5) *Integration*: Proximity to transit hubs (<200m)

Their taxonomy reveals a research gap: only 12% of studies incorporated infrastructure quality metrics into station siting decisions.

VI. CRITICAL GAP ANALYSIS AND RESEARCH AGENDA

A. Synthesized Gap Matrix

Appendix A consolidates limitations across the 12 surveyed works: Table II synthesizes the core contributions, methodologies, and limitations of the 12 seminal works analyzed in this survey. The table is organized by research domain to facilitate cross-study comparison and identification of complementary approaches for integrated system design. Key insights from this synthesis: • **Infrastructure-demand disconnect**: Only 3/12 papers ([1], [2], [5]) explicitly connect infrastructure quality to demand modeling, revealing a critical integration gap

- **Validation limitations**: 9/12 studies rely on single-city validation; cross-city transferability remains largely unaddressed
- **Operational integration**: Only 2 papers ([9], [10]) bridge prediction and optimization, yet without uncertainty propagation
- **Equity gap**: Only 1 paper ([3]) systematically addresses equity metrics; demographic accessibility remains underexplored

This comparison provides a foundation for designing integrated systems that simultaneously address infrastructure assessment, demand forecasting, and operational optimization—the central contribution of our unified framework in Fig. 1.

TABLE I: Critical Research Gaps Across Infrastructure Assessment, Demand Prediction, and Operations

Domain	Identified Gaps	Impact
Infrastructure Assessment	• OSM completeness disparities (urban vs. suburban) • Limited validation beyond field audits • No standardized bikeability index across cities	• Biased planning toward dense cores • Infrastructure quality uncertainty propagates to demand models
Demand Prediction	• High data requirements (>6 months history) • Poor generalization across cities • Black-box architectures limit planner trust	• New systems lack prediction capability • Models fail during regime shifts (events, construction)
Operations	• Prediction-optimization disconnect • No uncertainty quantification in rebalancing • Equity rarely incorporated in routing	• Suboptimal resource allocation • Marginalized communities underserved

B. Forward-Looking Research Directions

Based on gap analysis, we propose four high-impact research directions:

1. Multimodal Infrastructure Verification: Fuse mobile LiDAR, street view imagery, and OSM to create self-updating infrastructure maps. LiDAR boundary feature extraction [6] can detect missing bike lanes; computer vision can assess pavement quality closing the validation loop for open-data approaches. [13]

2. Cross-City Transfer Learning: Develop domain adaptation techniques where models trained on data-rich cities (e.g., Paris) transfer knowledge to emerging systems via infrastructure-aware graph alignment (matching station roles

rather than coordinates) and meta-learning for few-shot adaptation with <30 days of local data.

3. Uncertainty-Aware Operations: Integrate probabilistic forecasting (e.g., Bayesian ST-GCNs) with stochastic optimization to generate rebalancing plans robust to demand uncertainty critical for equitable service during anomalous events.

4. Equity-Centered Bikeability Metrics: Extend current infrastructure-focused metrics to incorporate demographic accessibility (service coverage by income/age groups), perceived safety (via social media sentiment analysis), and connectivity to essential destinations (schools, clinics, jobs).

VII. COMPARISON OF KEY APPROACHES

This synthesizes the core contributions, methodologies, and limitations of the 12 seminal works analyzed in this survey. The table is organized by research domain to facilitate cross-study comparison and identification of complementary approaches for integrated system design.

Key insights from this synthesis:

- **Infrastructure-demand disconnect:** Only 3/12 papers ([1], [2], [5]) explicitly connect infrastructure quality to demand modeling, revealing a critical integration gap
- **Validation limitations:** 9/12 studies rely on single-city validation; cross-city transferability remains largely unaddressed
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VIII. CONCLUSION

This survey synthesizes a fragmented research landscape into a cohesive framework where infrastructure quality, demand dynamics, and operational decisions form an interdependent system. Open data (particularly OSM) has democratized infrastructure assessment but requires multimodal verification to ensure reliability. Deep learning architectures (ST-GCNs, TFTs) have revolutionized demand forecasting yet remain disconnected from operational decision-making and suffer from poor cross-city generalization.

The most significant opportunity lies at the intersection of these domains: *infrastructure-aware demand models* that incorporate bikeability scores as graph edge weights, and *prediction-informed optimization* that propagates forecast uncertainty into rebalancing decisions. Future work must prioritize equity metrics and real-world deployment constraints to ensure cycling ecosystems serve all urban residents.

REFERENCES

- [1] J. Pucher and R. Buehler, “Cycling towards sustainability: Recent advances in cycling promotion policies,” *Transport Reviews*, vol. 43, no. 1, pp. 1–25, 2023, Available here.
- [2] C. Ferster, T. Nelson, K. Manaugh, J. Beirsto, K. Laberee, and M. Winters, “Developing a national dataset of bicycle infrastructure for Canada using open data sources,” *Transportation Research Part A: Policy and Practice*, vol. 179, p. 103945, 2024, Available here.
- [3] L. Wysling and R. S. Purves, “Where to improve cycling infrastructure? assessing bicycle suitability and bikeability with open data in the city of Paris,” *Journal of Transport Geography*, vol. 108, p. 103542, 2023, Available here.
- [4] Z. Yang, J. Chen, J. Hu, Y. Shu, and P. Cheng, “Mobility modeling and data-driven closed-loop prediction in bike-sharing systems,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 12, pp. 4488–4499, 2019, Available here.
- [5] N. V. Viero, S. T. Bech, N. Kaarsberg, N. B. Frölich, and C. S. Jensen, “Bikedna: A tool for bicycle infrastructure data and network assessment,” in *Proceedings of the 32nd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, 2024, pp. 112–123, Available here.
- [6] C. Li, R. Li, Y. Cai, and Y. Fu, “A lidar openstreetmap matching method for vehicle global position initialization based on boundary directional feature extraction,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 3, pp. 2105–2118, 2024, Available here.
- [7] I. Tomaras, K. Boutsis, and V. Kalogeraki, “Modeling and predicting bike demand in large city situations,” in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 4567–4578, Available here.
- [8] Z. Chen, H. Wu, N. E. O’Connor, and M. Liu, “A comparative study of using spatial temporal graph convolutional networks for predicting availability in bike sharing schemes,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 2, pp. 1892–1905, 2024, Available here.
- [9] M. Behrooz and A. Edrisi, “Bike sharing demand prediction using gated temporal graph convolutional networks,” *Transportation Research Part C: Emerging Technologies*, vol. 158, p. 104432, 2024, Available here.
- [10] B. Lim, S. Ö. Arik, N. Loeff, and T. Pfister, “Temporal fusion transformers for interpretable multi-horizon time series forecasting,” *International Journal of Forecasting*, vol. 40, no. 1, pp. 275–294, 2024, Available here.
- [11] M. Karunanithi, P. Chatasawapreeda, and T. A. Khan, “A predictive analytics approach for forecasting bike rental demand,” *Decision Analytics Journal*, vol. 11, p. 100482, 2024, Available here.
- [12] F. Cueva, P. Shi, and P. Cedillo, “Designing bike sharing systems supported by data: A systematic literature review,” *Transportation Research Record*, vol. 2678, no. 5, pp. 412–429, 2024, Available here.
- [13] E. O’Mahony and D. B. Shmoys, “Data analysis and optimization for (city)bike sharing,” *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, pp. 687–693, 2015, Available Here.
- [14] W. Zhang, P. Liu, and R. Chen, “Deep learning based bike sharing demand prediction using hybrid lstm and spatial features,” *IEEE Access*, vol. 12, pp. 15 432–15 445, 2024, Available Here.
- [15] J. Liang, M. C. M. Silva, D. Aloise, and S. D. Jena, “Dynamic rebalancing of bike sharing systems with time windows,” *EURO Journal on Transportation and Logistics*, vol. 12, no. 1, p. 100045, 2023, Available Here.

Appendix A: Synthesis of 12 Key Papers: Core Research Ideas, Methodologies, and Identified Gaps

Ref	Core Research Idea	Proposed Work	Methodology/Algorithms	Gaps in Research
[1]	Model cycling sustainability policies for urban mobility transformation	Comprehensive review of policy interventions promoting cycling adoption across global contexts	Policy taxonomy development; case study analysis; evaluation of co-benefits (health, emissions, equity)	Limited quantitative modeling of policy impacts; context-specific effectiveness; implementation barriers not fully addressed
[2]	Create consistent national cycling infrastructure dataset for Canada	OSM classification using Can-BICS scheme; systematic accuracy assessment across 15 cities	Rule-based OSM tagging classification into infrastructure type/comfort class; stratified random sampling; statistical accuracy estimation	OSM tagging ambiguity (bike paths vs. multi-use paths); imperfect accuracy; requires ongoing updates & contributor engagement
[3]	Identify street/city-level locations needing infrastructure improvement using open data	Reproducible pipeline fusing OSM + Apur data; validate 30 streets; produce suitability & bikeability maps	Data fusion of OSM network with Apur attributes; network accessibility modeling on grid; rule-based suitability scoring	Depends on OSM completeness/correctness; limited validation scope; transferability requires local destination definitions
[5]	Standardize bicycle infrastructure data preparation & network assessment	BikeDNA toolkit providing reproducible pipeline for data cleaning, metrics computation, output generation	Network graph processing; topology cleaning (self-loop removal, segment merging); quality metric computation (connectivity, directness)	Tool effectiveness depends on input data quality & local semantics; requires city-specific calibration/validation
[6]	Improve vehicle positioning via LiDAR-OSM feature matching	Boundary directional feature extraction from LiDAR matched to OSM map features for global position initialization	Feature extraction (directional boundary features); geometric matching algorithms; robustness to urban canyon effects	OSM feature quality/alignment errors affect matching accuracy; requires handling of sparse/incorrect map objects
[7]	Build predictive models for bike demand to support planning & operations	Demand forecasting framework anticipating spatiotemporal demand patterns instead of reactive operations	Historical usage analysis; feature engineering (temporal, weather, events); comparative evaluation of forecasting models	Poor generalization across cities; weak modeling of external factors (events, disruptions); limited interpretability
[14]	Predict bike demand by capturing both spatial dependencies & temporal trends	Hybrid deep learning model integrating LSTM with spatial correlation features + external covariates	Data preprocessing; spatial correlation mapping; hybrid LSTM architecture with attention-based fusion layer	High computation cost; requires large historical datasets; not tested across multiple cities; real-time deployment not addressed
[8]	Compare ST-GCN approaches for station availability prediction	Evaluate multiple ST-GCN variants on real BSS data to identify strengths/weaknesses per prediction horizon	Graph construction (spatial proximity + functional relationships); comparative ST-GCN evaluation (DCRNN, Graph WaveNet); ablation studies	Assumes fixed graph topology; struggles with missing data, station changes, demand shocks; limited real-world deployment validation
[9]	Improve demand prediction under spatiotemporal dependencies using gated mechanisms	Temporal GCN with gating units to dynamically weight historical observations based on temporal relevance	Gated temporal graph convolution; dynamic edge weighting; station graph construction with functional relationships	Limited robustness to graph drift during infrastructure changes; deployment complexity; interpretability challenges for planners
[10]	Forecast BSS availability using multi-horizon attention mechanisms	Temporal Fusion Transformer pipeline handling static, known, & observed inputs for interpretable forecasting	Variable selection networks; multi-horizon attention; static/dynamic covariate integration; interpretability via attention maps	High computational cost ($3.8\times$ ST-GCN); requires careful feature engineering; limited explainability for operational decisions
[12]	Systematically compile BSS design studies to extract key factors for strategic planning	SLR methodology covering 96 studies (2001–2023) identifying factors, methods, data sources	Systematic literature review with automatic + manual searches; screening/inclusion protocol; factor taxonomy development	Explicitly reports persistent gaps in long-term BSS design research; limited focus on infrastructure-demand integration
[15]	Optimize bike rebalancing operations under time window constraints	Vehicle routing formulation with time windows minimizing repositioning cost + demand mismatch penalty	Mathematical optimization (MILP); time-window constrained routing; integration with demand forecasts; simulation evaluation	Requires highly accurate demand forecasts; scalability challenges for large systems; limited handling of demand uncertainty